

CPS I — Exercise Sheet 8

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Exercise 1: Invariant checking I

$$U = \emptyset, \pi = \emptyset$$

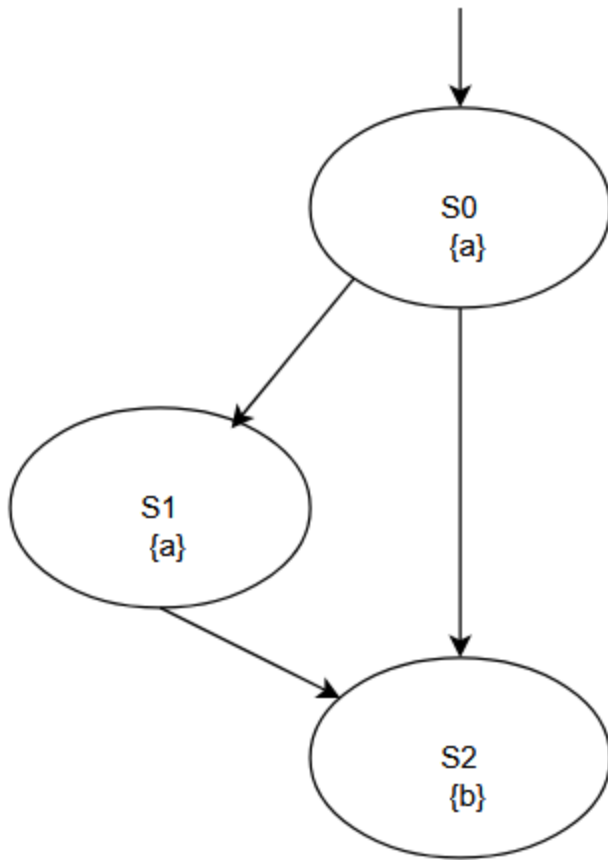
- $\text{DFS}(S_0, a) U = \{S_0\}, \pi = \{S_0\}$
 - $\text{DFS}(S_3, a) U = \{S_0, S_3\}, \pi = \{S_0, S_3\}$
 - $\text{DFS}(S_3, a) U = \{S_0, S_3\}, \pi = \{S_0, S_3, S_3\}$
 - $\text{DFS}(S_2, a) U = \{S_0, S_3, S_2\}, \pi = \{S_0, S_3, S_2\}$
 - $\text{DFS}(S_0, a) U = \{S_0, S_3, S_2\}, \pi = \{S_0, S_3, S_2, S_0\}$
- $\text{DFS}(S_1, a) U = \{S_1\}, \pi = \{S_1\}$
 - $\text{DFS}(S_3, a) U = \{S_1, S_3\}, \pi = \{S_1, S_3\}$
 - $\text{DFS}(S_3, a) U = \{S_1, S_3\}, \pi = \{S_1, S_3, S_3\}$
 - $\text{DFS}(S_4, a) U = \{S_1, S_4\}, \pi = \{S_1, S_4\}$
 - $\text{DFS}(S_3, a) U = \{S_1, S_4, S_3\}, \pi = \{S_1, S_4, S_3\}$
 - $\text{DFS}(S_3, a) U = \{S_1, S_4, S_3\}, \pi = \{S_1, S_4, S_3, S_3\}$

$$S_0, S_1, S_2, S_3 \models a$$

$$S_4 \not\models a$$

$S_0 \not\models \text{always } a$ counter example: S_1, S_4

Exercise 2: Invariant checking II



$$\Phi = a$$

The counterexample with non-minimal length S0,S1,S2

A counterexample of minimal length S0,S2

Exercise 3: Invariants

(a) No Idea

(b)

- $\sigma = (\{a\}\emptyset)^\omega$
 - For $A_i = \{a\}$, $\sigma = (\{a\}\{b\})^\omega$, $\sigma \in E$
 - For $A_i = \emptyset$, $\sigma = (\emptyset)^\omega$, $\sigma \in E$
- $\sigma = (\{a\}\{b\})^\omega$
 - For $A_i = \{a\}$, $\sigma = \{a\}^\omega$, $\sigma \in E$
 - For $A_i = \{b\}$, $\sigma = \{b\}^\omega$, $\sigma \in E$

- $\sigma = (\{a\}\emptyset)^\omega$
 - For $A_i = \{a\}$, $\sigma = \{a\}, \{a\}, \emptyset^\omega$, $\sigma \in E$
 - For $A_i = \emptyset$, $\sigma = \{a\}, \{a\}, \emptyset^\omega$, $\sigma \in E$